

Topic 1: Sets and Logic¹

THE LANGUAGE OF SETS

A **set** is a collection of distinct objects. Usually, we use the braces, $\{\}$, to represent the sets. The objects inside the braces are called **elements** of the set.

For example, here are some sets:

- $A = \{\text{peaches, kiwis, berries}\}$
- $S = \{0, 1, 2\}$
- $\emptyset = \{\}$
- $\mathbb{N} = \{1, 2, \dots\}$

A set that contains no elements at all, as in the third example, is called an **empty set** and is mathematically denoted as $\emptyset$.

Usually, we use upper capital letters to denote sets (e.g., S), and use lowercase letters to denote elements (e.g., x). To denote membership or inclusion in a set, we use the symbol $\in$. We denote “ x is an element of the set S ” with $x \in S$. For example, $\text{peaches} \in A$. We can also denote “ x is *not* an element of the set S ” with $x \notin S$. For example, $\text{bananas} \notin A$.

Example 1

True or false:

- $0 \in \{0, 1, 2\}$
- $0 = \{0\}$
- $\emptyset \in \emptyset$
- $\emptyset \in S$
- $S \in S$
- $S \in \mathbb{N}$

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In set theory, a set can be defined in two ways:

- **Extensionally** (by enumeration), by listing all of its elements.
- **Intensionally** (by description), by describing a property that defines its members.

On the previous examples, we used extensional definitions, listing each element within the sets. However, we can also define a set intensionally, using a condition to describe its elements, as illustrated below:

$$\{x \mid x \text{ has property } P.\}$$

means the collection of all elements that have the property P .

Sometimes you may also see notations like

$$\{x \in A \mid x \text{ has property } P.\},$$

which is equivalent to

$$\{x \mid x \in A \text{ and } x \text{ has property } P.\}.$$

Here we list some set notations that are commonly used.

$\mathbb{N}$	Set of natural numbers $\{1, 2, 3, \dots\}$. Some textbooks also include 0.
$\mathbb{Z}$	Set of integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$
$\mathbb{R}$	Set of real numbers (whole number line)
$\mathbb{R}^+$ or $\mathbb{R}_+$	Set of positive real numbers. Some textbooks define it as non-negative.
(a, b)	The open interval between real numbers a and b $(\{x \in \mathbb{R} \mid a < x < b\})$
$[a, b]$	The closed interval between real numbers a and b $(\{x \in \mathbb{R} \mid a \leq x \leq b\})$

Example 2

Rewrite the follow sets as listing all the elements:

- $\{x \in \mathbb{Z} \mid x \text{ is an even number.}\}$
- $\{x \geq 0 \mid x \text{ is an even number.}\}$
- $\{p \in \mathbb{Z} \mid p > 10 \text{ and } p < 2\}$

Definition 1 (Subsets)

For any two sets A and B , we say A is a **subset** of B if every element of A is also an element of B . We denote it as $A \subset B$ (or $A \subseteq B$).

If $A \subset B$ and $B \subset A$, we say A and B are **equal sets** ($A = B$).

If $A \subset B$ but $B \not\subset A$, we say A is a **proper subset** of B ($A \subsetneq B$).

Two sets are **equal sets** if they contain exactly the same elements. We write $A = B$ whenever $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$.² By definition, $A \subset B$ implies that every element of A also belongs to B ; that is, $x \in A \Rightarrow x \in B$. Therefore, two sets are equal if and only if each is a subset of the other: $A = B$ if and only if $A \subset B$ and $B \subset A$.

Example 3

True or false:

- $\{0, 2\} \subset \{x \geq 0 \mid x \text{ is an even number.}\}$
- $\{0, 2\} \subset \{x > 0 \mid x \text{ is an even number.}\}$
- $\{0, 2\} \subset \{0, 2\}$
- $\emptyset \subset \{0, 2\}$
- $\emptyset \subset \emptyset$

POWER SET

For any given set S , we can list all of its possible subsets. The collection of subsets of a set is called **the power set**.

Definition 2 (Power Set)

For any set S , **the power set** of the set S , denoted as $\mathcal{P}(S)$, is defined as the following:

$$\mathcal{P}(S) = \{A \mid A \subset S\}.$$

For example, the power set of $S = \{a, b, c\}$ is

$$\mathcal{P}(S) = \{\{\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

The empty set, $\emptyset$, is in the power set of any set as it is a subset of any set, including itself.

²The symbol $\Rightarrow$ denotes logical implication. A formal definition will be provided later in these notes.

CARDINALITY OF A SET

You can also count the number of elements in a set. The **cardinality** of the set A , denoted with $|A|$, is the number of elements in the set A .

If the number of elements in a set is finite, we call this set a **finite** set. If the number of the elements in a set is infinite, we call this set a **infinite** set.

However, there could be several levels of infinity. First, consider the set of all natural numbers, $\mathbb{N}$. It has infinitely many elements in it. Then we consider another set which has all non-negative integers, $\mathbb{Z}^+$. It seems that $\mathbb{Z}^+$ has more elements than $\mathbb{N}$. However, you can actually *count* the elements in $\mathbb{Z}^+$,

$$\mathbb{Z}^+ = \{ \begin{array}{cccc} 0, & 1, & 2, & 3, & \dots \end{array} \}$$

$$\begin{array}{cccc} \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} \end{array}$$

you can find a *one-to-one correspondence (bijection)* between the natural numbers and non-negative integers.³ Therefore we say $\mathbb{Z}^+$ has the same cardinality as $\mathbb{N}$ ($|\mathbb{Z}^+| = |\mathbb{N}|$), or $\mathbb{Z}^+$ is **countable** (or **countably infinite**).

You can even show that $\mathbb{Z}$ is also countable by indexing the elements in $\mathbb{Z}$ in the following alternating way:

$$\mathbb{Z} = \{ \begin{array}{cccccc} \dots & -2, & -1, & 0, & 1, & 2, & \dots \end{array} \}$$

$$\begin{array}{cccccc} \mathbf{5} & \mathbf{3} & \mathbf{1} & \mathbf{2} & \mathbf{4} \end{array}$$

Although not rigorously proven here, all subsets of $\mathbb{Z}$ and $\mathbb{Q}$ are also countable. However, $\mathbb{R}$ is uncountable.⁴

We can also consider the cardinality of the power set. If S is a finite set with n elements, then $\mathcal{P}(S)$ contains 2^n elements. That is, $|\mathcal{P}(S)| = 2^{|S|}$. In the previous section's example, we defined $S = \{a, b, c\}$, then $\mathcal{P}(S)$ has $2^3 = 8$ elements.

³We will review bijection in a few days.

⁴For the proof of the uncountability of $\mathbb{R}$, check *Cantor's diagonal argument* for reference.

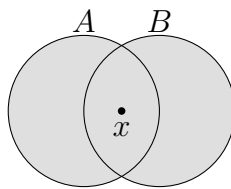
SET OPERATIONS

Just like that you can add or subtract numbers, you can also perform operations on sets.

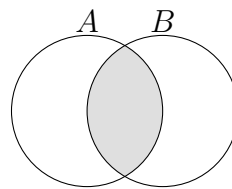
Definition 3 (Set operations)

- Union: $A \cup B := \{x \mid x \in A \text{ or } x \in B\}$.
- Intersection: $A \cap B := \{x \mid x \in A \text{ and } x \in B\}$.
- Difference: $A \setminus B$ or $A - B := \{x \mid x \in A \text{ and } x \notin B\}$.

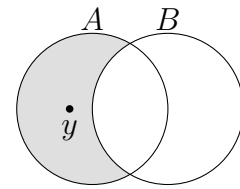
The figures below demonstrate these operations. These figures are called **Venn diagrams**. We use Venn diagrams to illustrate the relation between the sets. For example, in the left diagram in the first row, x is in the circles of both sets A and B , which represents that $x \in A$ and $x \in B$. Similarly, y is in the circle of set A but not in set B , which represents $y \in A \setminus B$ (right diagram). Finally, the intersection is represented in the central diagram. Note that when $A \cap B = \emptyset$, we say A and B are **disjoint sets** (as shown in the left figure in the second row).



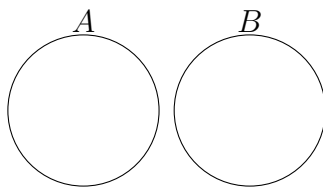
Union: $A \cup B$



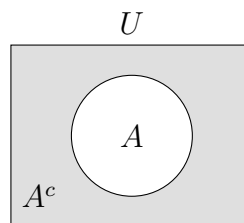
Intersection: $A \cap B$



Difference: $A \setminus B$



Disjoint Sets



A and A^c

We can define a **universal set** U that contains all the elements of interest. Then the **complement** of the set A , A^C or $\overline{A}$, is defined as

$$A^C = U \setminus A = \{x \mid x \in U, x \notin A\}.$$

We can take unions or intersections of more than two sets. We usually use the following notation.

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \cdots \cup A_n, \quad \bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$

The definition for the union or the intersection of multiple sets is similar to the two-set case. For example,

$$A \cup B \cup C = \{x \mid x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

We can also take unions or intersections of infinitely many sets.

Proposition 1 presents some useful facts that you may try to verify by yourself.

Proposition 1 (Distributive and associative property of set operations)

For any sets A , B , and C ,

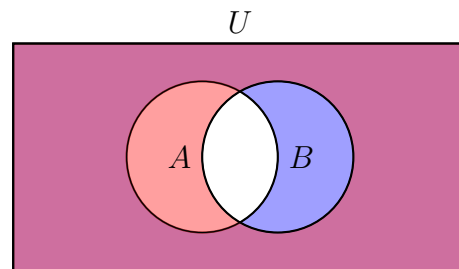
1. $A \setminus B = A \cap B^C$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive property)
3. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive property)
4. $A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C)$ (Associative property)
5. $A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$ (Associative property)

Proposition 2 (De Morgan's Law)

For any two sets A and B ,

- (1) $(A \cup B)^C = A^C \cap B^C$.
- (2) $(A \cap B)^C = A^C \cup B^C$.

We will now launch our first proof in this course. The first step of showing something, is to understand *what is going to be shown*. In the first part, we will show that the sets $(A \cup B)^C$ and $A^C \cap B^C$ are equal to each other. You may want to draw a Venn diagram to “prove” the proposition. Indeed, if you draw it, you will likely find that the proposition looks correct.



De Morgan's Law

However, drawing a Venn diagram is NOT a legit proof, especially if you just draw one. The proposition requires the condition *for any* two sets A and B . Drawing only one Venn diagram shows the one and only one very specific case that is represented with the graph. Nevertheless, graphing is still a very nice way to give yourself a quick glance at what the proposition is talking about and sometimes a hint about how you should proceed the proof.

How can we rigorously show the two sets are **equal**? You can try going back to the definition above: for any two sets A and B , $A = B$ if $A \subset B$ and $B \subset A$. Then we can establish the goal: to show that the two sets are subsets of each other.

Proof. We firstly show $(A \cup B)^C \subset A^C \cap B^C$. Pick $x \in (A \cup B)^C$. By definition, $x \notin A \cup B$. From this, we can know that $x \notin A$ and $x \notin B$, because if $x \in A$ or $x \in B$, x must belong to $A \cup B$ by definition. Hence, $x \in A^C$ and $x \in B^C$. By definition, it implies $x \in A^C \cap B^C$. Note that it is always true for any x that we pick. Therefore, $(A \cup B)^C \subset A^C \cap B^C$.

Then we show $A^C \cap B^C \subset (A \cup B)^C$. Pick $x \in A^C \cap B^C$. By definition, $x \in A^C$ and $x \in B^C$, which implies $x \notin A$ and $x \notin B$. Hence, $x \notin A \cup B$, so $x \in (A \cup B)^C$. Note that it is always true for any x that we pick. Therefore, $A^C \cap B^C \subset (A \cup B)^C$.

Since $(A \cup B)^C$ and $A^C \cap B^C$ are subsets of each other, $(A \cup B)^C = A^C \cap B^C$. ■

The above is the style of proof that you might read in textbooks, which is sometimes not the easiest to understand. You may also try the style of proof presented below.

Proof. WTS (want to show): $(A \cup B)^C \subset A^C \cap B^C$.

$$\begin{array}{ll}
 x \in (A \cup B)^C & \text{(take any } x \in (A \cup B)^C \text{)} \\
 \Rightarrow x \notin A \cup B & \text{(definition of complements)} \\
 \Rightarrow x \notin A \text{ and } x \notin B & \text{(by contradiction and definition of the union)} \\
 & \text{(if } x \in A \text{ or } x \in B, x \in A \cup B.) \\
 \Rightarrow x \in A^C \text{ and } x \in B^C & \text{(definition of complements)} \\
 \Rightarrow x \in A^C \cap B^C & \text{(definition of intersection)} \\
 \Rightarrow (A \cup B)^C \subset A^C \cap B^C & \text{(definition of subsets).}
 \end{array}$$

■

The other part of the proof is omitted. One thing to note is that: the former statement should always imply the latter statement when you write arrows. We will elaborate more when we talk about logic and proofs.

PARTITIONS OF A SET

Just like cutting a cake, you can cut a set into a collection of several subsets. This collection of subsets is called a **partition** of the set.

Definition 4 (Partition)

A collection of non-empty sets $\mathcal{P}$ is called a **partition** of a set S if it satisfies the following conditions:

- (1) For every set $A \in \mathcal{P}$, $A \subset S$
- (2) If $A \neq B$ and $A, B \in \mathcal{P}$, then $A \cap B = \emptyset$.
- (3) $\bigcup_{A \in \mathcal{P}} A = S$.

Partitions are useful in probability theory. For example, how can we calculate the probability of the event “getting two Heads when tossing two coins”? We can split the set of all possible events into partitions where each part has the same probability to happen, and *count* what’s the proportion that the “two-Heads” event happens in the partition. This is known as the **frequentist probability**.⁵

Example 4

Consider $S = \{1, 2, 3, 4\}$.

- Possible partition 1: $\mathcal{P}_1 = \{\{1\}, \{2\}, \{3\}, \{4\}\}$
- Possible partition 2: $\mathcal{P}_2 = \{\{1, 3\}, \{2, 4\}\}$
- Possible partition 3: $\mathcal{P}_3 = \{\{1, 2, 4\}, \{3\}\}$

Note that the partitions of the set S , in Example 4, are all subsets of the power set of S .

⁵You can also give a *measure* to the elements in a partition. We can define a **probability space** with respect to a partition (sigma algebra) and some valid probability measure.

CARTESIAN PRODUCT

The elements in a set are *unordered*. That is, the order in which we list elements does not change the essence of the set. However, there are situations when we need to consider mathematical objects with a specific order.

For example, let $x \in X = \{\text{labor, capital}\}$, $y \in Y = \{\text{output A, output B}\}$, and x produces y . We can have sentences like “labor produces output A” or “capital produces output B,” but not “output A produces labor”.

In the preceding example, the tuple (x, y) is an **ordered** pair; the order of the elements has meaning. The Cartesian product is used to describe the collection of such ordered pairs.

Definition 5 (Cartesian product)

Let X and Y be two sets. The **Cartesian product** of the two sets, $X \times Y$ (read as “ X cross Y ”), is defined as

$$X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}.$$

If $X = Y = \mathbb{R}$, then $X \times Y = \mathbb{R}^2$, the 2-dimensional Cartesian coordinate system.

In general, $X \times Y$ does not equal $Y \times X$, as seen in the above case. (Preference) relations are one of the most important applications of Cartesian products in economics.

Example 5

Let X be the set of the goods that can be chosen. For any $x \in X$ and $y \in X$, we write $x \succsim y$ if x is preferred to y . We can use the tuple (x, y) to express $x \succsim y$. The set of all such tuples can be expressed with the Cartesian product $\succsim \subset X \times X$.

a) If $X = \{a, b, c\}$, find $X \times X$.

We know Moona has the following preference:

$$a \succsim_M b, a \succsim_M c, b \succsim_M c,$$

then we can write $(a, b) \in \succsim_M$, $(a, c) \in \succsim_M$, and $(b, c) \in \succsim_M$.

Suppose Nina has the following preference:

$$a \succsim_N a, a \succsim_N b, a \succsim_N c, b \succsim_N b, b \succsim_N c, c \succsim_N c.$$

b) Find the relation $\succsim_N$. Is $\succsim_N = \succsim_M$?

ORDERED SETS**Definition 6** (Total Order)

Let S be a set. A relation $\leq$ defined on S is a **total order** if:

- (1) Reflexivity: $x \leq x$ for any $x \in S$.
- (2) Antisymmetry: if $x \leq y$ and $y \leq x$, then x and y are the same.
- (3) Transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$.
- (4) Completeness: Either $x \leq y$ or $y \leq x$.

Definition 7 (Ordered Set)

A set S is a **(total) ordered set** if there is a total order defined on S .

The notation “less than or equal to” ($\leq$) is commonly used to represent the ordering relation in an ordered set. It naturally captures the idea of one element being less than or the same as another, which aligns with the intuitive understanding of order.

Example 6

Convince yourself that the set of real numbers $\mathbb{R}$ is a total ordered set. The set of rational numbers $\mathbb{Q}$ is also a total ordered set.

The game of Rock-Paper-Scissors does not form an ordered set. Which property in the definition of total order is violated?

Definition 8 (Boundedness)

Suppose S is an ordered set and $E \subset S$.

If there exists $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is **bounded above**. We call β an **upper bound** of E .

If there exists $\alpha \in S$ such that $x \geq \alpha$ for every $x \in E$, we say that E is **bounded below**. We call α a **lower bound** of E .

Definition 9 (Supremum: Least Upper Bound)

Suppose S is an ordered set, and $E \subset S$. Suppose E is bounded above.

If there exists $\underline{\beta} \in S$ with the following properties:

- (1) $\underline{\beta}$ is an upper bound of E , and
- (2) for every upper bound of E , $\underline{\beta} \leq \beta$

Then $\underline{\beta}$ is called the **least upper bound** of E , or the **supremum** of E , and we write $\underline{\beta} = \sup E$.

Definition 10 (Infimum: Greatest Lower Bound)

Suppose S is an ordered set, and $E \subset S$. Suppose E is bounded below.

If there exists $\bar{\alpha} \in S$ with the following properties:

- (1) $\bar{\alpha}$ is a lower bound of E , and
- (2) for every lower bound of E , $\alpha \leq \bar{\alpha}$

Then $\bar{\alpha}$ is called the **greatest lower bound** of E , or the **infimum** of E , and we write $\bar{\alpha} = \inf E$.

Notice that we only require $\sup E$ and $\inf E$ to be in S , so $\sup E$ and $\inf E$ may or may not be in E .

Example 7

Let $E \subset \mathbb{R}$ consist of all numbers $\frac{1}{n}$, where $n = 1, 2, 3, \dots$

Convince yourself:

- (1) 1000 is an upper bound of E .
- (2) -1000 is a lower bound of E .
- (3) 1 is the least upper bound of E , which is in E .
- (4) 0 is the greatest lower bound of E , which is not in E .

AXIOMS

An **axiom** is a statement that is “taken to be true.” It serves as a starting point for further reasoning and arguments. Reasoning based on axioms is fundamental in microeconomic theory.

For example, suppose we want to prove that “I strictly prefer A to B” and “I strictly prefer B to A” cannot hold at the same time. We have the intuition that they cannot hold at the same time, but how can we formally prove this intuition? In microeconomic theory, we first propose a set of axioms that arguably “make sense.” Then, we show that when these axioms are true, these two statements cannot hold at the same time. Writing proofs based on a given set of axioms is essential in Econ 210A.⁶

Example 8

In Euclidean geometry, There’s a few axioms where all the properties can be derived with:

- (1) A straight line can be drawn between any two points;
- (2) A finite straight line can be extended indefinitely;
- (3) A circle can be drawn with any centre and any radius;
- (4) All right angles are equal;
- (5) The Parallel Postulate.

Elliptic Geometry replace the fifth axiom with: No parallel lines exist. Interesting properties are derived from this difference in axiom.

⁶In other fields of mathematics, the most common sets of axioms include the Peano axioms and the Zermelo–Fraenkel Choice (ZFC) axioms.

LOGIC

Logic is the field of study that determines the truthfulness and relations between statements. In order to analyze the statements, we need to break the statements into the atomic units: propositions. Here we start with the simplest case, that a proposition P can either be **true** or **false**. We say “ P has the truth value of T ” if P is true, and we say “ P has the truth value of F ” if P is false.

We can use a **truth table** to express the truth values of the proposition. For propositions P and Q , we can have truth values as follows.

P	Q
T	T
T	F
F	T
F	F

Since P and Q can either be true or false, there are in total four combinations of the truth states.

LOGICAL OPERATORS

Here we list some logical operators.

1. $\neg$ – Negation: $\neg P$, read as “not P .”
2. $\wedge$ – Conjunction: $P \wedge Q$, read as “ P and Q .”
3. $\vee$ – Disjunction: $P \vee Q$, read as “ P or Q .”
4. $\Rightarrow$ – Implication: $P \Rightarrow Q$, read as “ P implies Q ” or “if P , then Q .”⁷
5. $\Leftrightarrow$ – Equivalence: $P \Leftrightarrow Q$, read as “ P and Q are logically equivalent.”

The result of applying a logical operator to propositions is itself a proposition, so we can determine its truth value. Below are the truth tables for propositions involving these operators.

$$^7 P \Rightarrow Q \equiv \neg P \vee Q$$

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

You may find that the truth table somewhat reflects how we read these operators. For example, $\neg P$ is true only when P is false; $\neg P$ is false only when P is true. Similarly, $P \wedge Q$ is true when and only when both P and Q are true; if either P or Q is false, $P \wedge Q$ is false.

Note that $P \vee Q$ may contradict our daily use of **or**.⁸ We usually mean either P or Q are true, but not both. In computer science, this is called **exclusive or**, **strong or**, or **xor**.

Example 9

For any statements P , Q , and R , show that each group of statements below has the same truth table.

- (1) $\neg(\neg P)$ v.s. P (Double Negation)
- (2) $\neg(P \wedge Q)$ v.s. $\neg P \vee \neg Q$ (De Morgan's Law)
- (3) $\neg(P \vee Q)$ v.s. $\neg P \wedge \neg Q$ (De Morgan's Law)
- (4) $P \Rightarrow Q$ v.s. $\neg P \vee Q$ v.s. $\neg(P \wedge \neg Q)$ (Implication)
- (5) $P \Rightarrow Q$ v.s. $\neg Q \Rightarrow \neg P$ (Contrapositive)
- (6) $(P \wedge Q) \Rightarrow R$ v.s. $P \Rightarrow (Q \Rightarrow R)$ (Exportation)
- (7) $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ v.s. $P \Leftrightarrow Q$
- (8) $(P \Leftrightarrow Q) \wedge (Q \Leftrightarrow R)$ v.s. $P \Leftrightarrow R$ (Transitivity of equivalence)

IMPLICATION

Implication is the center of mathematical arguments. We usually use the implications to derive the conclusions we demand from the assumption that we know. In a statement $P \Rightarrow Q$, P is usually called the **premise** or the **assumption**, and Q is called the **conclusion**.

We can translate the following sentences into $P \Rightarrow Q$.

⁸“Would you like coffee or tea?” “Yes. ”

If P , then Q .
 Q if P .
 P only if Q .
 P implies Q .
 P is sufficient for Q .
 Q is necessary for P .

First, note that “if” and “only if” represent the complete opposite directions of the implication. Also notice the last two sentences above: when $P \Rightarrow Q$ is true, we call P the **sufficient condition** of Q , and Q the **necessary condition** of P .

You might find that the truth table for the implication is not very intuitive. Note that $P \Rightarrow Q$ is only false when P is true and Q is false. Statements such as “If a square has three sides, then the moon is made of cheese” or “Isla Vista is the largest city in the U.S. implies that the GDP of the U.S. in 2023 is decreased by 5%” both are true statements. Indeed, *any false premise implies any conclusion* is a true statement.

The key here is that we are finding whether the occurrence of P implies Q . P and Q are the evidences that may reveal this implication. If P is not true, we cannot prove that the implication $P \Rightarrow Q$ is false. In that case, as the implication is not proven false, it is true.

Example 10

Suppose P , Q , and R are statements. Use the truth table to show that the following statements are always true.

- (1) $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$ (*modus ponens*)
- (2) $((P \Rightarrow Q) \wedge \neg Q) \Rightarrow \neg P$ (*modus tollens*)
- (3) $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$ (*syllogism*)

If you explain *Modus ponens* in words, it says “when you know P is true, and you know if P then Q , Q is hence true”. In fact, this is how an argument works. On the other hand, *syllogism* provides the “chain” of arguments. These two rules form the basis of direct proof.

In Example 9, we also find rules related to implications. Specifically, (5) shows that $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ share the same truth table. We call $\neg Q \Rightarrow \neg P$ the **contrapositive** of $P \Rightarrow Q$, and $Q \Rightarrow P$ its **converse**. While a statement is logically equivalent to its contrapositive, which is important in mathematical arguments, it is not, in general, equivalent to its converse.

LOGICAL EQUIVALENCE

When the statements P and Q always have the same truth value, we say P and Q are **logically equivalent**, denoted as $P \Leftrightarrow Q$ or $P \equiv Q$. The pairs of statements we saw in Example 9 are logically equivalent.

From (7) in Example 9, we see that when both $P \Rightarrow Q$ and $Q \Rightarrow P$ hold, P and Q are logically equivalent. To indicate this, we use “ P if and only if Q ”, often abbreviated as “**iff**”.

The following sentences can be translated into $P \Leftrightarrow Q$.

- P if and only if (iff) Q .
- P is equivalent to Q .
- P characterizes Q .
- P is a sufficient and necessary condition for Q .
- P is defined as Q .

COMPOUND PROPOSITIONS, TAUTOLOGIES, AND CONTRADICTIONS

A **compound proposition** is a proposition formed by combining statements using logical operators. Depending on how it is formulated, a compound proposition may always be true or always false.

A compound statement that is always true is called a **tautology**, whereas one that is always false is called a **contradiction**.

For example, for any proposition P ,

$$P \vee (\neg P)$$

is always true, while

$$P \wedge (\neg P)$$

is always false.

The statements in Example 10 are all tautologies. Do not confuse tautology with logical equivalence.

OPEN SENTENCES AND QUANTIFIERS

Some statements cannot be determined true or false until they are completed. For example,

$$P(x) : x \geq 2.$$

We would not know whether this statement is true or false until we know which x we are talking about. When $x = 2$, it is true; when $x = 1$, it is false. This type of statement $P(x)$ is called an **open sentence**.

To determine the truth value of an open sentence, we need to specify the elements to be inserted into the sentence.⁹

Definition 11 (Some Quantifiers)

- $\forall$ – Universal quantifier
“ $\forall x \in X, P(x)$ ” is true if $P(x)$ is true for every $x \in X$.
- $\exists$ – Existential quantifier
“ $\exists x \in X$ such that $P(x)$ ” is true if there exists an $x \in X$ such that $P(x)$ is true.
- $\exists!$ – Uniqueness existential quantifier
“ $\exists! x \in X$ such that $P(x)$ ” is true if there exists **one and only one** $x \in X$ such that $P(x)$ is true.

Sometimes an open sentence can be a tautology. That is, no matter which x is inserted, it is always true. For example,

$$P(x) : x^2 \geq 0$$

is always true for any $x \in \mathbb{R}$ (although not necessarily true for some $x \in \mathbb{C}$).

Here is a useful property when dealing with the quantifiers. Carefully read this statement.

Proposition 3

If $P(x)$ is an open sentence with variable x , and X is a set, then

- (1) $\neg(\forall x \in X, P(x)) \Leftrightarrow \exists x \in X$ such that $\neg P(x)$.
- (2) $\neg(\exists x \in X$ such that $P(x)) \Leftrightarrow \forall x \in X, \neg P(x)$.

⁹If X is not specified, we consider all x in the universe U .

In general, the statements switching quantifiers, that is,

$$\forall x, \exists y \text{ such that } P(x, y) \quad \text{and} \quad \exists y \text{ such that } \forall x, P(x, y)$$

are not equivalent.

We can verify this by an example: for any integer x , there is some integer y that is larger than x . That is,

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \text{ such that } y > x \quad \text{and} \quad \exists y \in \mathbb{Z} \text{ such that } \forall x \in \mathbb{Z}, y > x$$

are not equivalent, where the former is true while the latter is not.

You may feel that the statements with quantifier has a lot of similarities to the set language. Indeed, we can rewrite the statements with sets. For example, consider the statement $\forall x, P(x) \Rightarrow Q(x)$. Let

$$P = \{x \mid P(x) \text{ is true}\}, \quad Q = \{x \mid Q(x) \text{ is true}\}.$$

By definition, P is a subset of Q if and only if every element in P belongs to Q . Symbolically,

$$P \subset Q \Leftrightarrow (\forall x, x \in P \Rightarrow x \in Q) \Leftrightarrow (\forall x, P(x) \text{ is true} \Rightarrow Q(x) \text{ is true}).$$

When properly defined, the logical statements can all be represented by the set language.

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